

Machine Vision Course Project

Automatic License Plate Recognition (ALPR) System – Enhancement & Evaluation Project

1. PROJECT OVERVIEW

In this project, you will design, implement, and evaluate an **Automatic License Plate Recognition (ALPR)** system. You may use any existing ALPR framework, open-source pipeline, or pretrained models available online. However, your work **must introduce an innovative improvement** in at least one component of the ALPR pipeline, such as:

- License plate detection
- Character segmentation
- Character recognition
- Pre-processing / post-processing enhancements
- Data augmentation strategy
- Model training improvements
- Robustness methods for environmental variations
- Any novel idea that improves accuracy, speed, or generalization

Your goal is to understand ALPR concepts thoroughly and critically evaluate the performance of your system.

2. REQUIREMENTS

A. ALPR System Development

You must:

1. Select and justify the baseline ALPR system you are using.
2. Implement at least **one meaningful innovation** to improve performance.
3. Prepare your own training/validation pipeline OR enhance an existing training approach.
4. Use a suitable dataset (public or self-collected). Discuss:
 - Dataset source & description
 - Characteristics (lighting, angles, plate variations)
 - Limitations & biases
 - Pre/post-processing, augmentation steps performed (if any)
 - Provide complete code on **GitHub**. README must contain installation steps and usage instructions.

B. PERFORMANCE EVALUATION

1. On **Week 12**, you will receive a **test dataset** consisting of Malaysian vehicle plate images captured under:
 - Indoor/outdoor lighting
 - Day/night
 - Different CCTV cameras
 - Angled / blurred / noisy conditions
 2. During the **Week 14 presentation**, you will:
 - Run your ALPR system live on the test dataset
 - Report performance (accuracy, character accuracy, detection success, etc.)
 - Explain failure cases & discuss possible improvements
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3. Deliverables (Due Week 14)

1. Presentation + Live Demo

You will present your system, architecture, innovation, dataset, and results. A live demo using the Week 13 test dataset is compulsory.

2. GitHub Repository

Your GitHub repo must include:

- All source code
- Pre/post-processing scripts
- Model files or instructions to download them
- A clear README with:
 - Setup instructions
 - How to run the ALPR system
 - Explanation of your methodology
- Link to the one-page poster

3. One-Page Poster

Prepare a compact and visually clear **one-page poster** (PowerPoint, Canva) containing:

1. **Introduction**
 - What is ALPR and why it matters
 - The problem statement
2. **Methodology**
 - Baseline ALPR approach
 - Your proposed improvements
 - Pipeline diagram
3. **Results & Discussion**
 - Test dataset accuracy (character-level / plate-level)
 - Visual examples
 - Strengths & weaknesses
4. **Conclusion**
 - Summary of key findings
 - Possible future improvements
5. **References & GitHub Link**
 - Cite all tools, models, papers, and code you adapted
 - Provide the GitHub repository link

Upload the poster to your GitHub repo.

Marking Scheme (20 Marks)

Component	Marks
Technical Implementation & Innovation	6
Performance on Test Dataset	5
Presentation & Understanding	5
GitHub Code & Documentation	2
One-Page Poster	2
Total	20